Lecture 7 — Exercises

Advanced QML: Gradients & Barren Plateaus

UZH Quantum Machine Learning Course — Dr. Pavel Sulimov, ZHAW

Prerequisites: basic PQC notation, PennyLane familiarity

The exercises are ordered by difficulty: **A** (warm-up), **B** (core), **C** (advanced). Starred items (★) are optional challenge problems for fast finishers. All code exercises assume Python 3.10+ with `pennylane` $\geq$ `0.44`. Use `pennylane.numpy` (not standard numpy) for circuit parameters.

1. Warm-Up: Parameter-Shift Rule by Hand

Exercise A1 — Manual gradient with parameter-shift

Consider the 1-qubit circuit

$$|\psi(\theta)\rangle = R_Y(\theta)|0\rangle, \qquad R_Y(\theta) = e^{-i\frac{\theta}{2}Y} = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

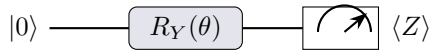
with observable $O = Z$.

(a) Write down $f(\theta) = \langle\psi(\theta)|Z|\psi(\theta)\rangle$ as a closed-form function of θ.

(b) Compute $f'(\theta)$ analytically (classical calculus).

(c) Apply the parameter-shift rule: $f'(\theta) = \dfrac{f(\theta + \frac{\pi}{2}) - f(\theta - \frac{\pi}{2})}{2}$. Verify that this gives the same result as (b).

(d) Evaluate both expressions numerically at $\theta = \pi/4$. What are the values of f, f'?



2. Hybrid Gradient: Chain Rule

Exercise A2 — Hybrid classical-quantum gradient

A hybrid model consists of:

$$z = wx + b \quad \longrightarrow \quad \text{circuit: } \langle Z\rangle(\theta)|_{\theta=z} \quad \longrightarrow \quad \hat{y} = v\langle Z\rangle$$

with scalar loss $\mathcal{L} = (\hat{y} - y_*)^2$.

(a) Draw the computation graph, labelling each node.

(b) Derive $\partial\mathcal{L}/\partial w$ using the chain rule. Your answer should involve the parameter-shift evaluation $\partial\langle Z\rangle/\partial\theta$.

(c) With $x = 1$, $w = 0.5$, $b = 0.2$, $v = 1$, $y_* = 0.8$ and $\eta = 0.1$, perform **one gradient step** updating w.

Hint

$\partial\mathcal{L}/\partial w = 2(\hat{y} - y_*) \cdot v \cdot \frac{\partial\langle Z\rangle}{\partial\theta} \cdot x$; remember $\theta = wx + b$, so $\partial\theta/\partial w = x$.

3. Core: Computational Cost Analysis

Exercise B1 — Cost of gradient computation

A PQC has $n = 6$ qubits and $L = 3$ layers, each containing one R_Y, one R_Z, and one CNOT on every qubit pair. Assume only R_Y and R_Z are parametrised.

(a) Count the total number of trainable parameters p.

(b) How many circuit executions does one gradient step require under the **parameter-shift rule**?

(c) If each execution uses $S = 1000$ shots, how many measurements does one full gradient step cost?

(d) If we replace parameter-shift with **SPSA**, how many circuit executions are needed? What is the trade-off?

(e) Suppose training requires $T = 500$ gradient steps. Complete the table below:

Method	Runs/step	Total runs	Total measurements
Param-shift			
SPSA			
Finite diff.			

4. Core: Barren Plateau Diagnosis

Exercise B2 — Detecting barren plateaus experimentally

You are given the following empirical gradient variance data from random initialisations across different qubit counts:

n (qubits)	$\hat{\sigma}^2$ (model A)	$\hat{\sigma}^2$ (model B)
4	0.050	0.048
6	0.012	0.031
8	0.003	0.021
10	7×10^{-4}	0.013
12	2×10^{-4}	0.009

(a) Plot $\log(\hat{\sigma}^2)$ vs. n for both models on the same axes (can be hand-drawn).

(b) Fit a straight line to each series. What does the slope tell you?

(c) Compute the **plateau severity index** $\mathcal{P}(n) = 1 - \hat{\sigma}^2(n)/\hat{\sigma}^2(n_0)$ with $n_0 = 4$ for $n = 12$. Which model is more affected?

(d) One model suffers from a barren plateau; the other does not. Identify which is which and suggest one architectural change that could improve the problematic model.

Hint

If $\hat{\sigma}^2 \propto 2^{-n}$, then $\log_2 \hat{\sigma}^2$ decreases linearly with slope -1. A shallower slope suggests only polynomial suppression (or a local cost function was used).

5. Core: Coding — PennyLane Barren Plateau Demo

Exercise B3 — Coding: observe barren plateaus in PennyLane

Use `lecture7_tasks.ipynb` for the full interactive version. A skeleton is reproduced here for reference. Requires PennyLane ≥ 0.44.

```
# Skeleton | fill in the TODOs
import pennylane as qml
import pennylane.numpy as np  # required for requires_grad=True
import matplotlib.pyplot as plt

def make_circuit(n_qubits, n_layers):
    dev = qml.device("default.qubit", wires=n_qubits)

    @qml.qnode(dev, diff_method="parameter-shift")
    def circuit(params):
        # TODO: implement a hardware-efficient ansatz:
        # - L layers of RY(params[l,q]) + RZ(params[l,q + n_qubits]) on each
        ↪  qubit q
        # - CNOT entanglement chain: qubit 0->1->...->n-1 after each layer
        ...
        return qml.expval(qml.PauliZ(0))  # global cost

    return circuit

qubit_counts = [2, 4, 6, 8, 10]
n_layers = 3
n_samples = 200
variances = []

for n in qubit_counts:
    circ = make_circuit(n, n_layers)
    grads = []
    for _ in range(n_samples):
        params = np.array(np.random.uniform(0, 2*np.pi, (n_layers, 2*n)),
                          requires_grad=True)
        # TODO: compute gradient w.r.t. params[0, 0] using qml.grad
        # and append to grads
        ...
    variances.append(np.var(grads))

# TODO: plot log(variance) vs n_qubits and interpret
```

(a) Complete the skeleton. Run it and plot the results.

(b) What slope do you observe in the $\log_2(\text{variance})$ vs. n plot?

(c) Modify the cost function to use a **local** observable $\sum_q \langle Z_q \rangle / n$ instead of $\langle Z_0 \rangle$. Re-run and compare the variance scaling.

Bonus / Challenge

Replace `diff_method="parameter-shift"` with `diff_method="spsa", num_directions=1`. Measure wall-clock time for one gradient step at $n = 10$ for both methods. What speedup do you observe?

6. Advanced: Escaping Barren Plateaus

Exercise C1 — Identity-block initialisation

The **identity-block** strategy initialises parameters so that each layer of the PQC implements approximately the identity map. For a layer of $R_Y(\theta_q)$ gates followed by CNOT chain:

(a) What parameter values make $R_Y(\theta)$ equal to the identity?

(b) Show that at this initialisation, the gradient $\partial \mathcal{L} / \partial \theta_k$ is *not* exponentially small (argue qualitatively or via a 2-qubit example).

(c) Implement identity-block initialisation in the skeleton from Exercise B3. Add small Gaussian noise $\epsilon \sim \mathcal{N}(0, 0.01)$ to break symmetry. Compare variance with random initialisation.

Hint

$R_Y(\theta) = \mathbb{I}$ when $\theta \in \{0, 2\pi, \ldots\}$. Near the identity, the circuit is in a "product state region" and expectation values depend strongly on each local gate.

Exercise C2 — Rotosolve: analytic single-parameter optimisation

For a 1-parameter rotation gate, the loss landscape is exactly:

$$\mathcal{L}(\theta) = A \sin(\theta) + B \cos(\theta) + C$$

for some A, B, C that depend on the rest of the circuit.

(a) Using three evaluations $\mathcal{L}(0)$, $\mathcal{L}(\pi/2)$, $\mathcal{L}(-\pi/2)$, derive closed-form expressions for A, B, C.

(b) Find the optimal θ^* that minimises $\mathcal{L}(\theta)$.

(c) Implement the Rotosolve step for a single parameter in Python (no PennyLane autodiff needed — just three circuit evaluations).

(d) Apply Rotosolve cyclically to all parameters of the 6-qubit circuit from B1 for 100 cycles. Plot the loss vs. cycle. How does convergence compare to gradient descent?

Exercise C3 — Critical thinking: the expressibility–trainability trade-off

(a) In your own words, explain why a *more expressive* PQC is harder to train. Use the concentration-of-measure intuition.

(b) Suppose you are designing a QML model for a 12-qubit problem. You have two circuit families:

- **Family X**: 4 layers, full entanglement, 96 parameters
- **Family Y**: 2 layers, nearest-neighbour CNOT, 48 parameters

Without running any code, which do you expect to suffer more from barren plateaus? Justify your answer.

(c) Propose a training strategy that lets you use Family X (for its expressibility) while mitigating barren plateaus. Be concrete: list at least 3 techniques in order of application.

7. Mini-Project (Take-home)

Mini-Project: Barren Plateau Mitigation Benchmark

Goal: empirically compare 3 barren-plateau mitigation strategies on a fixed 8-qubit hardware-efficient ansatz for a VQE-style task (minimise $H = \sum_i Z_i Z_{i+1} + \sum_i X_i$, the transverse-field Ising model).

Strategies to compare:

1. Random initialisation + standard gradient descent (baseline)
2. Identity-block initialisation + gradient descent
3. Local cost function + gradient descent

Metrics to report:

- Gradient variance at initialisation vs. n (scaling plot)
- Loss convergence curve (loss vs. gradient steps)
- Final energy achieved after 300 steps
- Wall-clock time per gradient step

Deliverable: A short report (max 2 pages) with plots and a 1-paragraph discussion. Submit code as a Jupyter notebook.

Suggested reading: Cerezo et al. (2021), *Nat. Commun.* 12, 1791; Holmes et al. (2022), *PRX Quantum* 3, 010313.

Solutions (Instructor Copy — Do Not Distribute)

Solution sketch (instructor copy)

A1 — Parameter-shift rule

(a) $f(\theta) = \langle 0|R_Y^\dagger Z R_Y|0\rangle = \cos\theta$ (b) $f'(\theta) = -\sin\theta$

(c) $\frac{1}{2}[\cos(\theta+\pi/2) - \cos(\theta-\pi/2)] = \frac{1}{2}[-\sin\theta - \sin\theta] = -\sin\theta$ ✓

(d) At $\theta = \pi/4$: $f = 1/\sqrt{2} \approx 0.707$, $f' = -1/\sqrt{2} \approx -0.707$

Solution sketch (instructor copy)

A2 — Hybrid chain rule

(b) $\dfrac{\partial \mathcal{L}}{\partial w} = 2(\hat{y} - y_*) \cdot v \cdot \dfrac{\partial \langle Z\rangle}{\partial \theta} \cdot x$, where $\theta = wx + b$ and $\partial\theta/\partial w = x$.

(c) With given values: $\theta = 0.5{\cdot}1{+}0.2 = 0.7$, $\langle Z\rangle = \cos(0.7) \approx 0.765$, $\hat{y} = 0.765$.

PSR: $\frac{\partial\langle Z\rangle}{\partial\theta} = \frac{1}{2}[\cos(0.7{+}\pi/2) - \cos(0.7{-}\pi/2)] = -\sin(0.7) \approx -0.644$.

Gradient: $2(0.765{-}0.8)(1)(-0.644)(1) \approx 0.045$.

Update: $w_{\text{new}} = 0.5 - 0.1{\cdot}0.045 \approx 0.4955$.

Solution sketch (instructor copy)

B1 — Computational cost

(a) $p = 2 \times 6 \times 3 = 36$ parameters (b) $2p + 1 = 73$ runs/step (c) 73 000 measurements/step

(e)

Method	Runs/step	Total runs ($T = 500$)	Total measurements
Param-shift	73	36 500	36.5M
SPSA	2	1 000	1.0M
Finite diff.	37	18 500	18.5M

Solution sketch (instructor copy)

B2 — Barren plateau diagnosis

(b) Model A slope ≈ -1 ($\log_2$) $\Rightarrow$ exponential suppression $\Rightarrow$ barren plateau. Model B slope $\approx -0.45 \Rightarrow$ polynomial, trainable.

(c) $\mathcal{P}_A(12) = 1 - 2{\times}10^{-4}/0.05 = 0.996$; $\mathcal{P}_B(12) = 1 - 0.009/0.048 = 0.81$. Model A is nearly fully suppressed.

(d) Fix: switch Model A to a local cost $\mathcal{L} = \sum_i \langle Z_i\rangle$, or use identity-block initialisation.

Solution sketch (instructor copy)

C1 — Identity-block init

(a) $R_Y(\theta) = \mathbb{I}$ when $\theta \in \{0, 2\pi, 4\pi, \ldots\}$.

(b) Near $\theta = 0$: the circuit maps $|0\rangle \to |0\rangle$ on all qubits, $\langle Z\rangle = 1$ exactly. A small perturbation ϵ changes $\langle Z\rangle$ by $\sim \epsilon$ (linear regime), giving gradient $\sim \mathcal{O}(1)$, not exponentially small.

(c) Use `params = np.random.normal(0, 0.01, shape)` — see `lecture7_solutions.ipynb`.

Solution sketch (instructor copy)

C2 — Rotosolve

(a) $C = \mathcal{L}(0)$; $A = [\mathcal{L}(\pi/2) - \mathcal{L}(-\pi/2)]/2$; $B = [\mathcal{L}(\pi/2) + \mathcal{L}(-\pi/2)]/2 - C$

(b) $\theta^* = \operatorname{atan2}(-A, -B) + \pi$ (global min of $A \sin\theta + B\cos\theta + C$)

(c)
```python
def rotosolve_step(circuit_fn, params, k):
    p = params.flatten().copy()
    def f(t):
        p2 = p.copy(); p2[k] = t
        return float(circuit_fn(p2.reshape(params.shape)))
    C = f(0)
    A = (f(np.pi/2) - f(-np.pi/2)) / 2
    B = (f(np.pi/2) + f(-np.pi/2)) / 2 - C
    p[k] = np.arctan2(-A, -B) + np.pi
    return p.reshape(params.shape)
```

References

McClean, J.R. et al. (2018). Barren plateaus in quantum neural network training landscapes. *Nature Commun.* 9, 4812.

Cerezo, M. et al. (2021). Cost function dependent barren plateaus in shallow parametrized quantum circuits. *Nature Commun.* 12, 1791.

Ostaszewski, M. et al. (2021). Structure optimization for parameterized quantum circuits. *Quantum* 5, 391.

Holmes, Z. et al. (2022). Connecting ansatz expressibility to gradient magnitudes and barren plateaus. *PRX Quantum* 3, 010313.